Evaluation of Scenario Description Language (SDL)

This study aims to validate our Scenario Description Language (SDL), a framework designed to capture data-intensive application scenarios by describing both the characteristics of the data and the transformations required to derive actionable knowledge.

Your task is to:

- Review the guidelines to learn how to model scenarios with our SDL.
- Select a scenario from your experience and model it with the SDL using our modeling tool.
- Complete this questionnaire to share your feedback on the experience of modeling the scenario with the SDL.

Your expert opinion is invaluable for assessing the language's effectiveness and usability.

The questionnaire is divided into three sections: the first gathers information about your professional background, the second addresses the characteristics of the scenario you modeled, and the third focuses on your evaluation of the language.

The estimated time to complete the questionnaire is 15-20 minutes.

All data will be collected anonymously and used solely for research purposes.

* Indica una domanda obbligatoria

Participant Profile

This section helps us understand your background and context for your feedback.

1. What is your current primary role? *

Contrassegna solo un ovale.

- ◯ Researcher/Academic
- ◯ Phd Student
- ◯ Software Architect
- ◯ Data Engineer / Data Architect
- ◯ Software Developer
- ◯ Altro: _____

2. How many years of professional experience do you have in modeling software architectures? (i.e., the process of defining a structured solution that meets technical and operational requirements.)

Contrassegna solo un ovale.

- ◯ 0 - 2 Years
- ◯ 3 - 5 Years
- ◯ 6 - 10 Years
- ◯ 11 - 15 Years
- ◯ 15+ Years

3. How knowledgeable do you feel about software architecture modeling? * (Please rate on a scale from 1 to 10, where 1 is Novice and 10 is Expert)

Contrassegna solo un ovale.

	1	2	3	4	5	6	7	8	9	10	
Nov	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Expert

4. How would you rate your familiarity with Model-Driven Engineering (MDE) * or Domain-Specific Languages (DSLs)?

   Contrassegna solo un ovale.

   - ◯ Beginner (limited or no experience)
   - ◯ Intermediate (have used them in some projects)
   - ◯ Expert (Use them regularly or have designed them)

5. How familiar are you with architectural concepts for data-intensive systems (e.g., Data Mesh, Lambda/Kappa Architecture, Cloud-Fog-Edge computing)? (Please rate on a scale from 1 to 10, where 1 is Not Familiar and 10 is Very Familiar)

   Contrassegna solo un ovale.

	1	2	3	4	5	6	7	8	9	10	
Not	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Very Familiar

6. How many years of professional experience do you have in data-intensive distributed systems? (i.e., systems designed to process and manage large volumes of data across multiple distributed nodes.)

   Contrassegna solo un ovale.

   - ◯ 0 - 2 Years
   - ◯ 3 - 5 Years
   - ◯ 6 - 10 Years
   - ◯ 11 - 15 Years
   - ◯ 15+ Years

7. How knowledgeable do you feel about data-intensive distributed systems? *
(Please rate on a scale from 1 to 10, where 1 is Novice and 10 is Expert)

Contrassegna solo un ovale.

	1	2	3	4	5	6	7	8	9	10	
Nov	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Expert

8. In which domains have you had direct experience with data-intensive scenarios? (Select all that apply and briefly give an example if possible.)

Seleziona tutte le voci applicabili.

- ☐ Analytics / Business Intelligence
- ☐ Artificial Intelligence / Machine Learning
- ☐ Internet of Things (IoT)
- ☐ Edge / Fog / Cloud Computing

Modeling Tools and Experience

Please list the modeling tools or languages you have used previously. For each, please indicate your proficiency and the primary context in which you used it.
You can add multiple tools by selecting “Yes” in the last question of this section.

9. Tool / Language (e.g., UML, ArchiMate)

_____

10. Proficiency

Contrassegna solo un ovale.

- Beginner
- Intermediate
- Advanced
- Expert

11. Primary context of use

12. Do you want to add another tool/language? *

Contrassegna solo un ovale.

- Yes *Passa alla domanda 9.*
- No

Part 2: Characteristics of the Scenario

This section collects details about the scenario you chose to model.

13. Please, upload the JSON of the scenario you generated

File inviati:

14. Briefly describe the scenario you modeled with SDL. What was its main purpose and what knowledge did it aim to extract from the data?

15. Are the producers in your scenario external systems and/or data sources? *

Contrassegna solo un ovale.

- External Systems
- Data Sources
- Both

16. Please, briefly describe them *

17. Are the consumers in your scenario external systems and/or data sources? *

Contrassegna solo un ovale.

- External Systems
- Data Sources
- Both

18. Please, briefly describe them *

19. How would you rate the complexity of interactions between components in your scenario? (Please rate on a scale from 1 to 10, where 1 is Very Simple and 10 is Very Complex.) *

Contrassegna solo un ovale.

	1	2	3	4	5	6	7	8	9	10	
Very	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Very Complex

20. (Optional) Please, motivate your answer

21. Did the scenario focus more on data processing logic, data storage/management, or both equally? *

Contrassegna solo un ovale.

- ◯ Processing Logic
- ◯ Storage / Management
- ◯ Both Equally

22. (Optional) Please, motivate your answer

Part 3: Evaluation of the Modeling Language

This section gathers your feedback on our SDL. Please answer based on the modeling task you performed.

Expressiveness and Completeness

23. How well does the language capture the key concepts of data-intensive scenarios? (Pleas rate on a scale from 1 to 10, where 1 is Very Poorly and 10 is Very Well.)

Contrassegna solo un ovale.

	1	2	3	4	5	6	7	8	9	10	
Very	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Very Well

24. (Optional) Please, motivate your answer

25. Which non-functional aspects are well-supported by SDL? Check All that apply *

Seleziona tutte le voci applicabili.

- [ ] Performance
- [ ] Scalability
- [ ] Reliability
- [ ] Security
- [ ] Data Quality
- [ ] Maintainability
- [ ] Altro: _____

26. Do you think SDL adequately supports functional aspects? (Please rate on a scale from 1 to 10, where 1 is Not adequate ad all and 10 is Fully Adequate.) *

Contrassegna solo un ovale.

	1	2	3	4	5	6	7	8	9	10	
Not	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Fully Adequate

27. (Optional) Please, motivate your answer

_____

_____

_____

_____

_____

28. While designing your scenario using the SDL, how often did you encounter concepts or requirements that were difficult or impossible to model? *

Contrassegna solo un ovale.

- Never
- Once or Twice
- A few times
- Many times

29. If yes, please describe them, including why it has been difficult and how you would have preferred to model them.

30. Are the provided constructs (e.g., structural and behavioral elements) sufficient and intuitive for representing your scenario? (Please rate on a scale from 1 to 10, where 1 is Not at All and 10 is Completely.)

Contrassegna solo un ovale.

	1	2	3	4	5	6	7	8	9	10	
Not	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Completely

31. (Optional) Please, motivate your answer

32. To what extent do you think this language is specific to the data-intensive domain compared to a general-purpose language like UML? (Please rate on a scale from 1 to 10, where 1 is ”Not specific at all” and 10 is ”Highly specialized and effective”.)

Contrassegna solo un ovale.

	1	2	3	4	5	6	7	8	9	10	
Not	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Highly Specialized

33. (Optional) Please, motivate your answer

Usability and Learnability

34. How easy was it to learn and start using the language? (Please rate on a scale from 1 to 10, where 1 is Very Difficult and 10 is Very Easy.) *

Contrassegna solo un ovale.

	1	2	3	4	5	6	7	8	9	10	
Very	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Very Easy

35. (Optional) Please, motivate your answer

36. How clear and understandable are the language's notations and diagrams? (Please rate on a scale from 1 to 10, where 1 is Very Unclear and 10 is Very Clear.) *

Contrassegna solo un ovale.

	1	2	3	4	5	6	7	8	9	10	
Very	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Very Clear

37. (Optional) Please, motivate your answer

38. Did you find any parts of the notation or language constructs to be ambiguous or open to multiple interpretations?

Seleziona tutte le voci applicabili.

- [ ] Unclear terminology
- [ ] Ambiguous symbols or diagrams
- [ ] Composition rules not precise
- [ ] Altro: _____

39. If yes, please indicate the nature of the ambiguities.

_____

_____

_____

_____

_____

Efficiency and Effectiveness

40. Approximately how much time did it take you to model your scenario (after reading the guidelines)? *

Contrassegna solo un ovale.

- ◯ Less than 30 minutes
- ◯ 30-60 minutes
- ◯ 1-2 hours
- ◯ 2-4 hours
- ◯ More than 4 hours

41. Which elements of SDL made modeling more efficient? *

Seleziona tutte le voci applicabili.

- [ ] Notation Simplicity
- [ ] Abstraction Level
- [ ] Reusability of Constructs
- [ ] Expressiveness of the Language
- [ ] Clarity of Semantics
- [ ] Consistency and Unambiguity
- [ ] Altro: _____

42. If you used your old way to model the current case - what is the time needed * to model using the your way?

Contrassegna solo un ovale.

- ◯ Less than 30 minutes
- ◯ 30-60 minutes
- ◯ 1-2 hours
- ◯ 2-4 hours
- ◯ More than 4 hours

43. Which aspects most influenced the time difference between your previous modeling approach and the use of SDL (e.g., number of steps, level of abstraction, granularity of actions, others)?

_____

_____

_____

_____

_____

44. Were there any modeling steps that felt particularly time-consuming or repetitive? *

Contrassegna solo un ovale.

- ◯ No
- ◯ Yes

45. If yes, how do you think repetitiveness could be reduced?

_____

_____

_____

_____

_____

Communication and Maintainability

46. Would you prefer to use SDL diagrams over UML/ArchiMate diagrams when communicating with stakeholders(e.g., other architects, developers, data scientists)? (Please rate on a scale from 1 to 10, where 1 is Absolutely Not and 10 is Absolutely Yes.)

Contrassegna solo un ovale.

	1	2	3	4	5	6	7	8	9	10	
Abs	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Absolutely Yes

47. (Optional) Please, motivate your answer

48. Imagine you have to modify the scenario you just modeled in six months. How easy do you think it would be to understand and change the model? (Please rate on a scale from 1 to 10, where 1 is Very Difficult and 10 is Very Easy.) *

Contrassegna solo un ovale.

	1	2	3	4	5	6	7	8	9	10	
Very	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Very Easy

49. (Optional) Please, motivate your answer

Overall Feedback

50. What are the main strengths of the SDL, in your opinion? *

Seleziona tutte le voci applicabili.

- [ ] Time Efficiency
- [ ] Easy to Learn
- [ ] Clear Communication Language
- [ ] High Level of Abstraction
- [ ] Expressiveness of Models
- [ ] Reusability of Constructs
- [ ] Consistency and Unambiguity
- [ ] Tool Support
- [ ] Scalability for Complex Architectures
- [ ] Support for Scenario Comparison and Analysis
- [ ] Altro: _____

51. What additional elements could improbe the efficiency of our modeling language?

_____

_____

_____

_____

_____

52. If SDL were available as a supported tool, how likely would you be to adopt it in your work? (Please rate on a scale from 1 to 10, where 1 is Not at All and 10 is Definitely Yes.) *

Contrassegna solo un ovale.

	1	2	3	4	5	6	7	8	9	10	
Not	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Definitely Yes

53. (Optional) Please, motivate your answer

54. Would you recommend this modeling approach to other practitioners or researchers in you field?

Contrassegna solo un ovale.

- Yes
- Maybe
- No

55. (Optional) Please, motivate your answer

56. How scalable do you think SDL would be for large, real-world scenarios? (Please rate on a scale from 1 to 10, where 1 is Not at All and 10 is Definitely Yes.)

Contrassegna solo un ovale.

	1	2	3	4	5	6	7	8	9	10	
Not	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Definitely Yes

57. (Optional) Please, motivate your answer

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